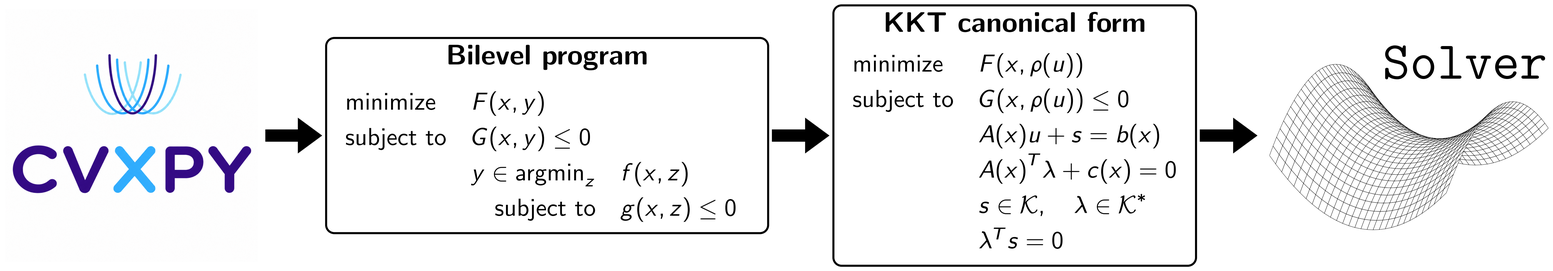
Disciplined Bilevel Programming

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Disciplined optimization

- constructive rules for building optimization problems
- makes problems easier to specify, analyze, and solve
- fast model prototyping without expert knowledge of optimization:
  Say what you want, not how to get it.
- initially proposed for convex optimization; extended to many nonconvex problems
- implemented as **domain specific languages** (DSLs) in MATLAB, Python, ...

Rulesets

$$\begin{aligned} &\text{minimize} && f(x;\omega) \\ &\text{subject to} && p_i(x;\omega) \sim q_i(x;\omega), \quad i = 1, \ldots, m \end{aligned}$$

- x: variables; ω: parameters; f: objective function
- p_i, q_i: constraint functions; '$\sim$' is one of $\{\leq, =\}$

disciplined convex program (DCP) [GBY06] compliant, if

- f convex (in x)
- p_i convex and q_i concave for '$\leq$'
- p_i and q_i affine for '$=$'

disciplined parameterized program (DPP) [AAB+19] compliant, if DCP when

- parameters ω treated as affine
- parameter-affine and parameter-free products treated as affine

disciplined nonlinear program (DNLP) [CZNB26] compliant, if

- f is linearizable convex
- p_i linearizable convex and q_i linearizable concave for '$\leq$'
- p_i and q_i smooth for '$=$'

Bilevel optimization problems

$$\begin{aligned} &\text{minimize} && F_0(x, y) \\ &\text{subject to} && F_i(x, y) \leq 0, \quad i = 1, \ldots, m \\ & && y \in S(x) \end{aligned} \tag{1}$$

where for a fixed x,

$$\begin{aligned} S(x) = \operatorname{argmin}_z \; & f_0(x, z) \\ \text{subject to } & f_i(x, z) \leq 0, \quad i = 1, \ldots, p \end{aligned} \tag{2}$$

- (1) called *upper problem*; (2) called *lower problem*
- x: *upper variable*; y: *lower variable*

disciplined bilevel program (DBLP) [ZB26] compliant, if

- $F_0, F_1, \ldots, F_m$ are DNLP-compliant
- $f_0, f_1, \ldots, f_p$ are DPP-compliant (so (2) is convex parameterized by x)

Single-level rewriting system

- DPP guarantees (2) is representable as conic KKT conditions

$$S(x) = \left\{ \rho(u) \;\middle|\; \begin{array}{ll} A(x)u + s = b(x), & A(x)^T\lambda + c(x) = 0 \\ s \in \mathcal{K}, & \lambda \in \mathcal{K}^*, \quad \lambda^T s = 0 \end{array} \right\} \tag{3}$$

  - u, s, λ are the variables; $\mathcal{K}$ and $\mathcal{K}^*$ are convex cone and its dual cone
  - canonicalization map $x \mapsto (A(x), b(x), c(x))$ is affine
  - solution recovery map ρ: $u \mapsto y$ is affine
  - both maps have fixed coefficients independent of values of x
- plug (3) into (1) returns a single-level DNLP-compliant problem
- solved via complementarity relaxation $\lambda^T s \leq \epsilon$ and continuation $\epsilon \to 0$

BLVPY

- available at `https://github.com/dxogrp/blvpy`
- DSL for bilevel programming in Python, built on top of CVXPY

Example: Ridge regression auto-tuning

consider the bilevel problem

$$\begin{aligned} &\text{minimize} && (1/m_{\text{val}})\|X_{\text{val}}w - y_{\text{val}}\|_2^2 \\ &\text{subject to} && 10^{-4} \leq \lambda \leq 10 \\ & && w \in S(\lambda) = \operatorname{argmin}_{\tilde{w}} (1/m_{\text{tr}})\|X_{\text{tr}}\tilde{w} - y_{\text{tr}}\|_2^2 + \lambda\|\tilde{w}\|_2^2 \end{aligned}$$

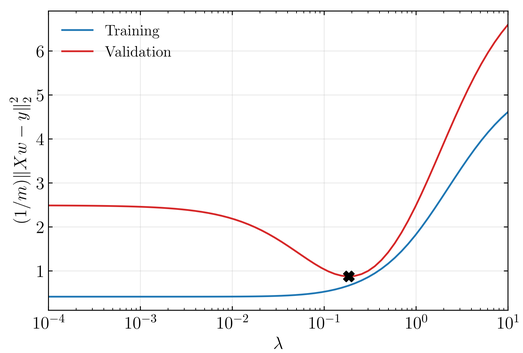
- upper variable: $\lambda \in \mathbf{R}$; lower variable: $w \in \mathbf{R}^n$
- finds the optimal regularization parameter λ that minimizes the validation error

to specify and solve this problem in BLVPY:

```python
import cvxpy as cp
import blvpy as bp

lbd, w = cp.Variable(nonneg=True), cp.Variable(n)
tr_loss = cp.sum_squares(X_tr @ w - y_tr) / m_tr
lower = bp.LowerProblem(
    cp.Minimize(tr_loss + lbd * cp.sum_squares(w)), parameters=[lbd],
)
val_loss = cp.sum_squares(X_val @ w - y_val) / m_val
prob = bp.BilevelProblem(
    cp.Minimize(val_loss), lower, [lbd >= 1e-4, lbd <= 10],
)
assert prob.is_dblp()
prob.solve()
```

results for $m_{\text{tr}} = 80$, $m_{\text{val}} = 24$, $n = 8$ (solution shown in black cross):



References

[AAB+19] A. Agrawal, B. Amos, S. Barratt, S. Boyd, S. Diamond, and J. Z. Kolter. Differentiable convex optimization layers. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[CZNB26] D. Cederberg, W. Zhang, P. Nobel, and S. Boyd. Disciplined nonlinear programming. *arXiv*, 2606.02896, 2026.

[GBY06] M. Grant, S. Boyd, and Y. Ye. Disciplined convex programming. In *Global Optimization: From Theory to Implementation*, pages 155–210. 2006.

[ZB26] H. Zhu and J. Boedecker. Disciplined bilevel programming. *arXiv*, 2609.00644, 2026.